

一、驱动篇

1.运行时候提示找不到master

```
hugo@hugo: ~/sdk/rs_driver/solution/rslidar_sdk1.5.10/build
File Edit View Search Terminal Help
=====
-- rs_driver Version : v1.5.10
=====
-- Configuring done
-- Generating done
-- Build files have been written to: /home/hugo/sdk/rs_driver/solution/rslidar_sdk1.5.10/build
Scanning dependencies of target rslidar_sdk_node
[ 33%] Building CXX object CMakeFiles/rslidar_sdk_node.dir/node/rslidar_sdk_node.cpp.o
[ 66%] Building CXX object CMakeFiles/rslidar_sdk_node.dir/src/manager/node_manager.cpp.o
[100%] Linking CXX executable rslidar_sdk_node
[100%] Built target rslidar_sdk_node
hugo@hugo:~/sdk/rs_driver/solution/rslidar_sdk1.5.10/build$ ./rslidar_sdk_node
*****
*****
***** RSLidar_SDK Version: v1.5.10 *****
*****
*****
[ERROR] [1691373713.266270446]: [registerPublisher] Failed to contact master at [localhost:11311]. Retrying...
```

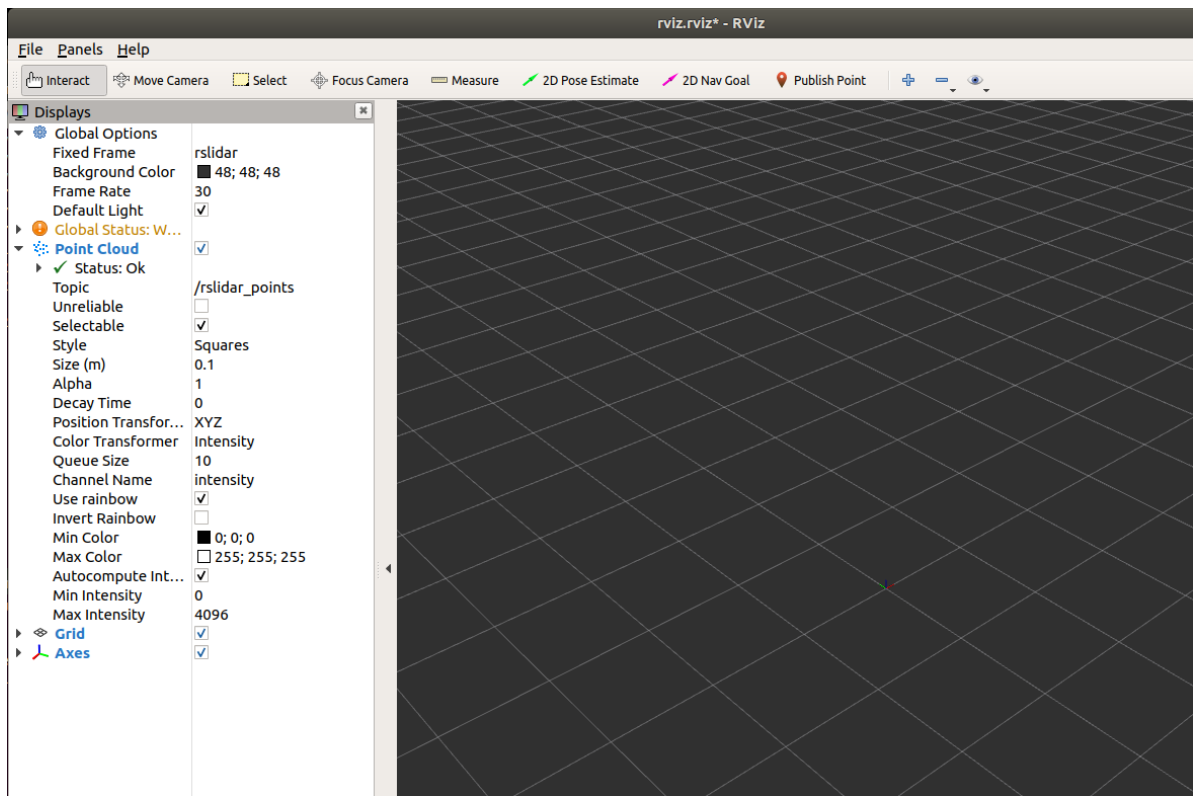
解决办法：开启新的终端，运行roscore

2.运行以后报黄色字体: ERRCODE_MSOPTIMEOUT

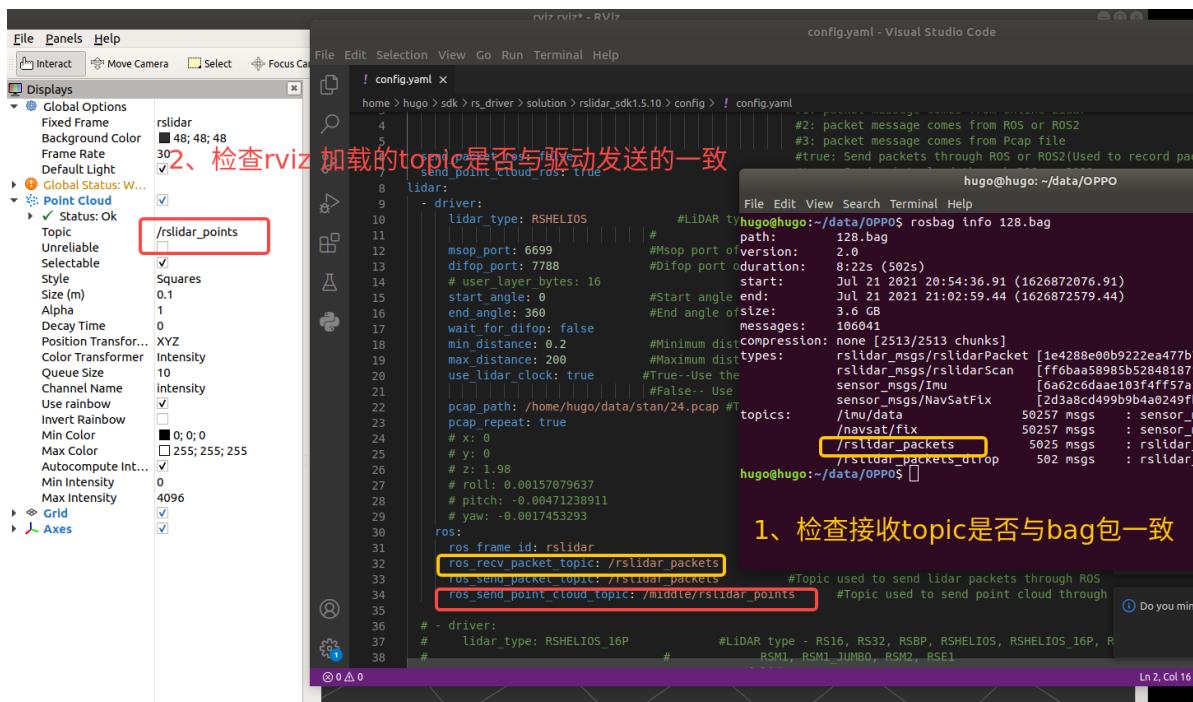
```
hugo@hugo: ~/sdk/rs_driver/solution/rslidar_sdk1.5.10/build
File Edit View Search Terminal Help
y: 0
z: 0
roll: 0
pitch: 0
yaw: 0
-----
Send PointCloud To : ROS
PointCloud Topic: /middle/rslidar_points
-----
RoboSense-LiDAR-Driver is running.....
ERRCODE_MSOPTIMEOUT
ERRCODE_MSOPTIMEOUT
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ERRCODE_MSOPTIMEOUT
```

解决办法：检查PC与雷达是否正常通信，或者是否存在网络阻塞

3.播放离线bag包的时候运行驱动，rviz没有点云显示



解决方法：1、检查接收的topic是否与bag包中的topic一致；2、检查rviz加载的topic是否与驱动发送的topic一致



4.播放pcap包时候提示错误

```
hugo@hugo:~/sdk/rs_driver/rslidar_sdk-1.3.2/build$ ./rslidar_sdk_node
*****
*****
***** RSLidar_SDK Version: v1.3.2 *****
*****
*****
-----
Receive Packets From : Pcap
Msop Port: 6699
Difop Port: 7788
-----
ERRCODE_PCAPWRONGPATH
Driver Initialize Error....
hugo@hugo:~/sdk/rs_driver/rslidar_sdk-1.3.2/build$
```

解决方法：检查pcap包的路径是否正确。

5.驱动其他问题请查询驱动使用说明手册

二、算法sdk篇

1.编译失败

```
hugo@hugo: ~/git/release/build
File Edit View Search Terminal Help
[ 2%] Building CXX object rs_tool/rs_beta/CMakeFiles/rs_beta.dir/map_filter/src
/external/map_filter.cpp.o
[ 2%] Building CXX object rs_tool/rs_beta/CMakeFiles/rs_beta.dir/src/rviz_displ
ay/beta_display.cpp.o
[ 2%] Building CXX object rs_tool/rs_beta/CMakeFiles/rs_beta.dir/src/perception
/beta_perception.cpp.o
[ 3%] Building CXX object rs_tool/rs_beta/CMakeFiles/rs_beta.dir/map_filter/src
/internal/mole_map_filter.cpp.o
[ 3%] Building CXX object rs_tool/rs_beta/CMakeFiles/rs_beta.dir/src/mechanic_s
ort/rs_mechanic_sort.cpp.o
[ 4%] Building CXX object rs_tool/rs_beta/CMakeFiles/rs_beta.dir/map_filter/src
/external/common/rs_hd_map.cpp.o
[ 5%] Building CXX object rs_tool/rs_dependence/CMakeFiles/rs_dependence.dir/sr
c/rs_dependence_manager.cpp.o
CMakeFiles/Makefile2:3209: recipe for target 'rs_tool/rs_beta/CMakeFiles/rs_beta
.dir/all' failed
make[1]: *** [rs_tool/rs_beta/CMakeFiles/rs_beta.dir/all] Error 2
make[1]: *** Waiting for unfinished jobs....
[ 5%] Linking CXX shared library librs_dependence.so
[ 5%] Built target rs_dependence
Makefile:129: recipe for target 'all' failed
make: *** [all] Error 2
hugo@hugo:~/git/release/build$ rm -rf *
```

问题原因：libs库使用错误，例如电脑是X86电脑，但是/release/libs中的perception库用的是arm的库

解决方案：替换正确的库

2.终端无打印，一直卡在ready处

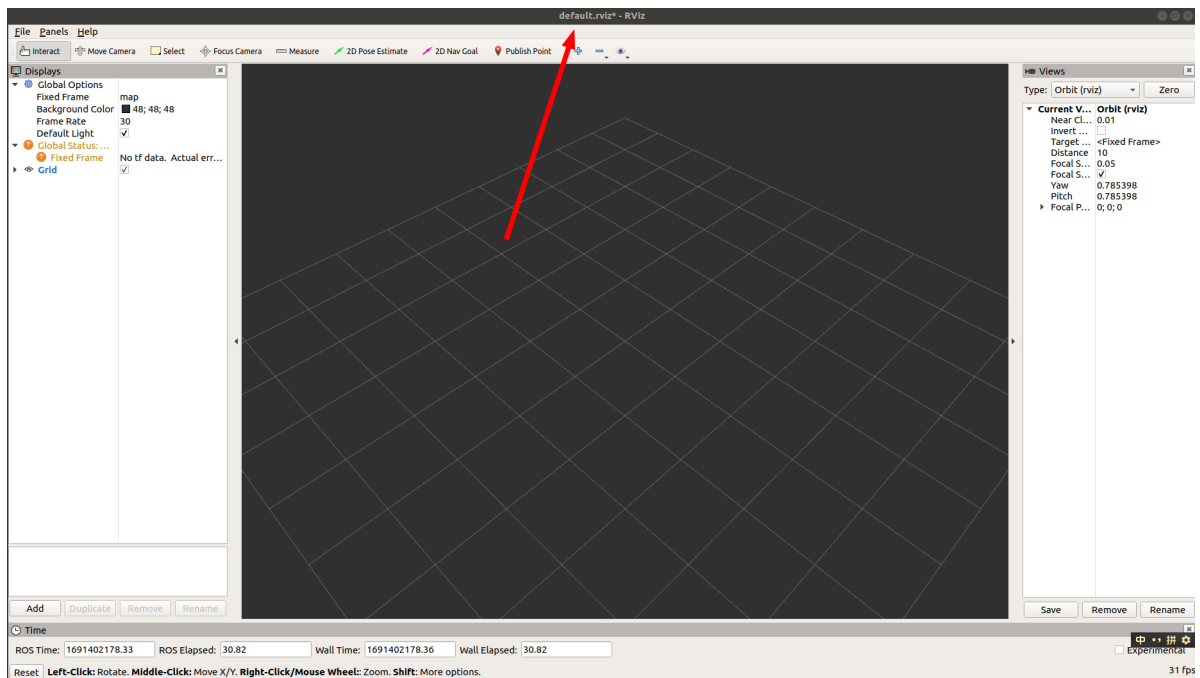
```
hugo@hugo: ~/git/release/build
File Edit View Search Terminal Help
Scanning dependencies of target receiver_demo
[ 98%] Building CXX object demo/CMakeFiles/rs_sdk_demo.dir/rs_sdk_demo.cpp.o
[100%] Building CXX object receiver_demo_beta/CMakeFiles/receiver_demo.dir/receiver_demo_beta.cpp.o
[100%] Linking CXX executable rs_sdk_demo
[100%] Built target rs_sdk_demo
[100%] Linking CXX executable receiver_demo
[100%] Built target receiver_demo
hugo@hugo:~/git/release/build$ ./demo/rs_sdk_demo
Checking authority of RS-Perception-X86...
Checking authority successfully!
/middle_lidar/PolariAiDetection: model version v1.0.0.R128.20201124
RsTrtInfer: load model and generate plan file ...
frame id /gnss
(x:0,y:0,z:0,roll:0,pitch:0,yaw:0)
frame id /imu
(x:0,y:0,z:0,roll:0,pitch:0,yaw:0)
frame id /middle_lidar
(x:0,y:0,z:0,roll:0,pitch:0,yaw:0)
frame id /odom
(x:0,y:0,z:0,roll:0,pitch:0,yaw:0)
RsApplication: version 3.1.0
RsApplication: ready!
```

问题原因：没有收到雷达数据

解决方案：1.在线雷达，检查与雷达连通性，与上述驱动检查一致；2.离线数据，检查bag包中的数据 topic 是否与usr_config.yaml文件中获取的topic一致

```
config > usr_config > ! usr_config.yaml
64 send_pos_ros: true # @type: bool @detail: if true, the localization algorithm result will be sent thr
65 send_pos_ros_topic: /rs_pose
66 send_fix_ros: true
67 send_map_ros: true
68 send_map_ros_topic: /rs_map
69 send_path_ros: true
70 send_path_ros_topic: /rs_path
71
72 # - method: Proto
73 # localization freq: 30
74 # send_pos_and_path: true
75 # socket:
76 # socket address: 10.10.8.239 # 对于发送
77 # socket port: 60802 # 对于接收
78 # socket buffer size: 4194304 # 默认即可
79 # max msg size: 32768 # 不允许大
80 # timeout ms: 150 # 接收超时，discouraged.
81 # send_control:
82 # send_control enable: true # hugo@hugo:~/data/OPPO$ rosbag info 128.bag
83 # send_control_thres: 202144 # path:
84 # send_control_ms: 3 # version: 2.0
85 # send_control_compress_enable: true # duration: 8:22s (502s)
86 # start: Jul 21 2021 20:54:36.91 (1626872076.91)
87 # end: Jul 21 2021 21:02:59.44 (1626872579.44)
88 # size: 3.0 GB
89 # messages: 186041
90 # compression: none [2513/2513 chunks]
91 # types:
92 # rslidar_msgs/rslidarScan [1e4288e0b9222ea477b73350bf24f51]
93 # rslidar_msgs/rslidarScan [ff6baa5898b5b528481871cbaf1bb342f]
94 # sensor_msgs/Imu [0a02c0daae1b3f4ff57a132d6f95cec2]
95 # sensor_msgs/NavSatFix [2d3ab0cd499bb04a0249fb98f8d0scfa40]
96 # /imu/data 50257 msgs : sensor_msgs/Imu
97 # /navsat/fix 50257 msgs : sensor_msgs/NavSatFix
98 # /rslidar_packets 5025 msgs : rslidar_msgs/rslidarScan
99 # /rslidar_packets_offset 502 msgs : rslidar_msgs/rslidarPacket
100
101 # hugo@hugo:~/data/OPPO$
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3.打开默认rviz无法显示感知结果



问题原因：没有添加对应的topic

解决方案：严格按照“快速启动手册”上写的，打开release/config/rviz中的perception.rviz

4.cube上（arm架构）下运行算法sdk，提示找不到.so文件

```

nvidia@nvidia-master:~/robosense/release/build/demo$ ./rs_sdk_demo
./rs_sdk_demo: error while loading shared libraries: libhasp_linux_arm64_25736.so: cannot open shared object file: No such file or directory
nvidia@nvidia-master:~/robosense/release/build/demo$

```

问题原因：ARM架构下需要声明一下libhasp_linux_arm64_25736.so文件所在路径

解决方案：运行rs_sdk_demo之前export LD_LIBRARY_PATH=(so文件所在路径):\$LD_LIBRARY_PATH，然后重新运行./rs_sdk_demo

5.运行算法提示key_not_found

```

hugo@hugo:~/sdk/3.1/release/build$ ./demo/rs_sdk_demo
Checking authority of RS-Perception-X86...
Checking authority Failed!
KEY_NOT_FOUND
hugo@hugo:~/sdk/3.1/release/build$

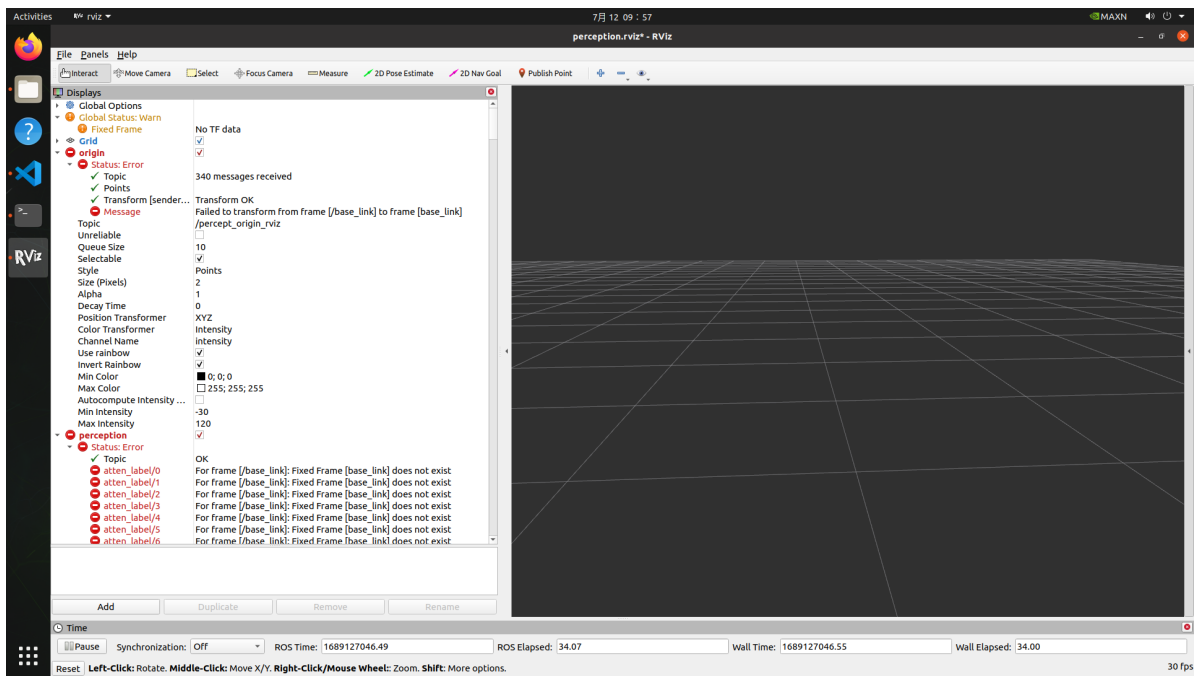
```

问题原因：1.没有插key；2.key过期了，或者是没有对应的权限；3.新电脑，没有装key驱动

解决方案：1.首先确认是否是key过期了，或者是适配X86的key用在了ARM架构上，又或者是key只有感知权限，而sdk把定位模块也打开了，具体Key权限和时限可以询问速腾的客户经理；2.如果是自己配的新电脑，需要安装key驱动：

```
sudo cp /release/80-hasp.rules /etc/udev/rules.d/
```

6.Ubuntu20.04上运行perception.rviz无法显示感知效果



问题原因：算法sdk针对Ubuntu20.04适配还有些小问题

解决方法：修改release/config/system_config/perception_config/rs_perception.yaml文件第53行 frame_id，将引号中"/base_link"中的"/"去掉，保存重新运行算法并重新打开perception.rviz文件即可。